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Nucleation and phase-selection in undercooled melts

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Abstract

Electromagnetic levitation technique is applied to undercool bulk melts of metals and alloys. A large undercooling range becomes accessible by avoiding heterogeneous nucleation on container walls and processing the melts under high purity environment. A freely suspended drop is produced with the extra benefit that it becomes accessible for direct observation and diagnostics. Temperature–time profiles (ttp) are measured during the melting and solidification process. The ttp profiles give insight into the thermodynamics of phase-selection processes. The levitation technique is combined with the diagnostic means at European Synchrotron Radiation Facility (ESRF Grenoble) to record time-resolved diffraction spectra during rapid crystallization of deeply undercooled melts. In such a way, the sequence of solidification of different crystallographic phases and (partly) remelting of primarily formed phases during recalescence is unambiguously determined. All these investigations allow for the construction of nucleation phase-selection diagrams examples of which are presented for Fe-based, Ni–V, and Nd–Fe–B alloys with a competition of phase formation of bcc, fcc, and some intermetallic phases, respectively. They give the conditions for the formation of metastable solids of different crystallographic structures as a function of concentration and undercooling achieved prior to solidification.
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Keywords: Undercooling; Containerless processing; Rapid solidification; Metastable phases; Hard magnetic alloys

1. Introduction

In the past, rapid quenching methods have been successfully applied to produce metastable solids from the liquid state. A great variety of metastable materials have been discovered, with physical properties, which make them suitable as new high-performance materials in mechanical and electrical engineering. To date, such metastable solids are even being produced on an industrial scale for their application in, e.g. steel production, design of electronic devices, and aerospace applications. In order to develop a predictive capability for modes of metastable solidification, efforts concentrate on the understanding, description, and modelling of the physical mechanisms [1] relevant to the formation of metastable phases from the liquid state. From a thermodynamic point of view, the undercooling of the melt is a necessary precondition for the solidification of metastable phases. If the nucleation temperature falls below the virtual melting temperature of a metastable solid a driving force for crystallization of this respective phase appears. Nucleation in the undercooled melt selects the crystallographic phase, stable or metastable before subsequent crystal growth completes solidification [2].

The present work reviews direct experimental investigations of nucleation and phase-selection in bulk undercooled melts of metals and alloys, and their impact on the formation of metastable materials. Containerless processing by electromagnetic levitation of bulk melts is applied for undercooling the melts essentially below their melting temperatures owing to the complete avoidance of container wall induced heterogeneous nucleation. A freely suspended drop gives the extra benefit that it becomes accessible for direct observation of rapid solidification, using proper diagnostic means. Even externally controlled nucleation of metastable phases becomes possible on freely levitated samples [3].

It is demonstrated that provided the melt is sufficiently undercooled prior to solidification, the nucleation of metastable bcc-phase in Fe–Ni is favored in competition with stable fcc-phase at Ni-concentrations larger than 4 at.% [4]. These investigations confirm the early experiments by Cech and Turnbull on small drop-tube processed droplets of Fe–Ni alloy [5]. The observation of a martensite-free bcc-phase upon solidification leads to the conclusion that the bcc-structure is primarily solidified as δ-phase from the undercooled melt

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[6]. Solidification pathways of liquid Fe–Ni samples processed containerlessly in drop-tubes are analyzed by microstructure investigations on small droplets as a function of concentration and droplet size [7]. Even in alloys of large Ni-content, nucleation of metastable bcc-phase is realized in levitation undercooling experiments by externally stimulating bcc solidification using a suitable nucleation trigger.

The preference for bcc nucleation over fcc nucleation is of special importance in the solidification of ferrite versus austenite phase in steel alloys and is experimentally proved in undercooled ternary Fe–Ni–Cr melts in the concentration regime where austenite is stable [9]. Ni–V alloy system is investigated in which, in addition to bcc and fcc solidification, the formation of intermetallic σ-phase also competes in the undercooled melt [10]. For each system a phase-selection diagram is constructed such that it gives quantitative values of composition and undercooling at which the primary crystallization of metastable bcc-phase is occurring in direct competition to other phases. The Nd–Fe–B alloy phase-selection during crystallization of the liquid is of direct practical relevance. Under equilibrium or near-equilibrium conditions the hard magnetic intermetallic Φ-phase $Nd_2Fe_{14}B$ is formed by a peritectic reaction $l + \gamma \rightarrow \Phi$ with α the soft-magnetic fcc-phase of Fe. Since peritectic reactions are in general very sluggish, Fe γ-phase remains and must be removed by time and energy-consuming processes in order to produce hard magnetic alloy containing Φ-phase [11]. It is demonstrated that undercooling is an efficient parameter in the primary solidification of hard magnetic Φ-phase from the melt [12].

The experiments and their results considered so far are based upon a careful analysis of the temperature–time profiles (ttp) measured during each undercooling experiment. High-frequency measurements of the rapid solidification during recalescence period (rapid temperature rise due to the release of the heat of crystallization) give reliable information on phase-selection by single or multiple recalescence events. However, they do not provide direct evidence of the formation of a metastable phase with a crystallographic structure different from that of the stable counterpart. Diffraction experiments on the as solidified alloys are often not useful because any metastable phase primarily formed may transform by either remelting during recalescence or solid-state transformation during the cooling period (small cooling rates, 10–100 K/s, in levitation experiments). Since the typical time scales of multiple recalescence events are in the order of a few seconds or even less, the use of conventional X-ray sources are of no help to record diffraction patterns during rapid solidification of undercooled melts. But the situation changes if high-intensity synchrotron radiation is taken into consideration. Therefore, electromagnetic levitation technique is combined with the means at the European Synchrotron Radiation Facility in Grenoble (France) to record diffraction patterns in time intervals of 0.5 s [13]. It is demonstrated that the sequence of phase-selection during rapid crystallization of undercooled melts of Ni–V alloys is unambiguously recorded and an experimentally determined phase-selection diagram of Ni–V is constructed [14].

2. Experimental

Samples having a mass of about 1 g were prepared from constituents of purity better than 99.99% by premelting in an induction furnace into spheres of about 6 mm diameter. Undercooling conditions were established by the application of the electromagnetic levitation technique. The experiments were performed under high purity environmental conditions. Temperatures were measured contactlessly by pyrometry. For further experimental details [2].

3. Experimental results and discussion

3.1. Activation energy of crystal nucleation

Nucleation into a specific crystallographic phase is characterized by an activation energy (ΔG^*) to form a nucleus of critical size in the undercooled melt. The nucleation barrier arises from the interfacial energy (σ_{sl}) between the crystal nucleus and the undercooled melt. According to classical nucleation theory [15] and to the negentropic model by Spaepen [16] for the estimation of σ_{sl}, ΔG^* is given by:

$$\Delta G^* = \frac{16\pi\sigma_{sl}^3}{3\Delta G^2} \cdot f(\theta) \quad \text{with} \quad \sigma_{sl} = \alpha \frac{\Delta S_f}{N_A^{1/3} \cdot V_m^{2/3}} \cdot T \quad (1)$$

where *f(θ)* is catalytic potency factor in the case of heterogeneous nucleation, ΔG^* is Gibbs free energy difference per unit volume between the solid and the liquid phase, ΔS_f is entropy of fusion, N_A is Avogadro's number, V_m is molar volume, T is temperature, and α is a factor depending on the structure of the nucleus with numerical values $\alpha = 0.71$ for bcc, and $\alpha = 0.86$ for fcc or hcp structures, respectively. Numerical solutions of the interface model by Spaepen also allow for calculations of α for crystallographic phases of other structures, e.g. intermetallic phases with large unit cells [17]. Hence, the barrier for nucleation, i.e. the interfacial energy σ_{sl}, depends on the structure of the nucleus. The interfacial energy is often determined by measuring the maximum undercooling attainable for a melt [18]. The Gibbs free energy difference is often determined by using the linear approximation, $\Delta G = \Delta S_f \cdot \Delta T$ proposed by Turnbull [19].

3.2. Phase-selection in Fe–Ni and Fe–Ni–Cr alloys

The dependence of the activation energy on the crystallographic structure of the nucleus has a strong impact on the solidification of metastable phases. As has been shown by studies on the model system Fe–Ni, a metastable bcc-phase crystallizes even in the concentration regime where the

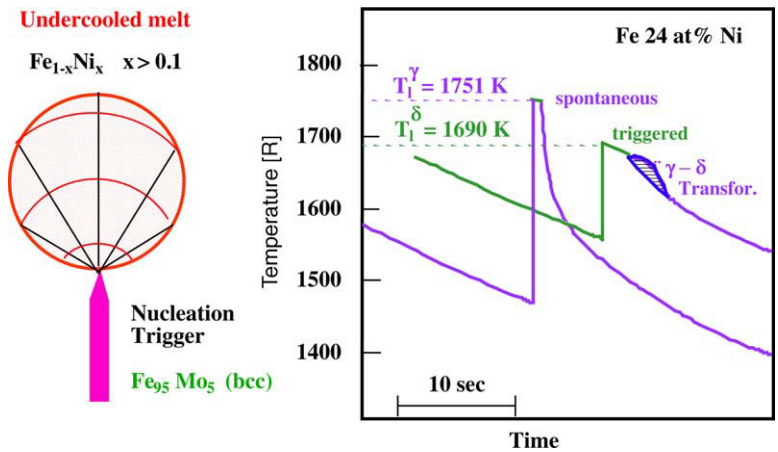

Fig. 1. Temperature–time profiles for spontaneous and externally triggered crystallization of undercooled $Fe_{76}Ni_{24}$ sample. During spontaneous crystallization stable fcc-phase is formed, but by nucleation triggering with a $Fe_{95}Mo_5$ needle of bcc-phase (cf. left side) the metastable bcc-phase in Fe–Ni alloy is successfully seeded. The primarily formed metastable bcc-phase transformes into stable fcc-phase during cooling as indicated by the hump on the respective temperature–time profile.

fcc-phase is stable, provided a critical undercooling is exceeded [4]. The critical undercooling for metastable bcc solidification rapidly increases with Ni-concentration so that at concentrations larger than 10 at.% Ni the achievable undercoolings are not sufficiently high for solidification of the metastable bcc-phase. But even in this concentration range metastable bcc crystallization is possible when using the method of externally stimulated heterogeneous nucleation of metastable bcc-phase. In this case the Fe–Ni melt is undercooled and subsequently touched by a nucleation trigger needle made of a $Fe_{95}Mo_5$ alloy. This material shows a stable high-temperature phase with a structure throughout similar to the metastable bcc-phase in Fe–Ni. In such a way the nucleation of bcc solidification is heterogeneously catalyzed in undercooled Fe–Ni melts [8]. Fig. 1 illustrates both the principle of the experimental procedure (left-hand side) and the temperature–time profiles recorded for spontaneous and triggered nucleation (right-hand side). The primary crystallization of metastable bcc-phase in $Fe_{76}Ni_{24}$ alloy is observed through a temperature rise during recalescence up to the virtual melting temperature of metastable bcc-phase, $T_l^\delta = 1690$ K. T_l^δ is lower than the liquidus temperature of stable fcc-phase, $T_l^\gamma = 1751$ K, up to which the temperature rises during spontaneous crystallization. The primarily formed metastable bcc-phase transforms into stable fcc-phase upon cooling of the as solidified sample as indicated by the small hump superimposed on the temperature–time profile in the case of triggered nucleation. As a consequence formation of the metastable bcc-phase cannot be checked by X-ray diffraction on these samples at ambient temperature.

According to Eq. (1) the results on metastable bcc-formation in Fe–Ni melts can be understood by a smaller activation energy to form critical nuclei of metastable bcc crystallites either through the smaller structural factor α in the interfacial energy as observed during spontaneous crystallization of deeply undercooled Fe-rich Fe–Ni alloys [4] or through a substantial reduction of the catalytic potency factor $f(\theta)$ for heterogeneous nucleation of bcc-phase in more concentrated Fe–Ni alloys due to external nucleation triggering with a material that shows stable bcc-phase at high temperatures [8].

Similar results on the formation of metastable bcc-phase have been obtained from analogous studies on the ternary alloy system Fe–Ni–Cr [20,9]. This alloy exhibits a phase competition between ferrite (bcc) and austenitic (fcc) phase, which is similar to that observed in the binary Fe–Ni system. However, it is of more practical relevance as the basic system for steel production.

The dependence of the formation of different crystallographic phases on undercooling obtained prior to solidification is used to develop phase-selection diagrams. Such diagrams correspond to phase diagrams, but contain, besides the concentration, the undercooling as a parameter for the formation of different crystallographic phases. As an example, Fig. 2 depicts the phase-selection diagram of the pseudo-binary alloy of isopleth $Fe_{69}Cr_{31-x}Ni_x$ [21].

The solid lines in the phase diagram represent the liquidus and the solidus temperatures as well as the boundary between the austenitic (fcc) and the ferrite (bcc) regime. The squares and triangles denote the primary nucleation of the bcc-phase and the fcc-phase, respectively. On the left-hand side (ferrite regime), bcc solidification is observed to be independent on undercooling. However, on the right-hand side (austenitic regime), metastable bcc solidification occurs at undercoolings exceeding a critical value, which changes with concentration. The results have been analyzed by calculating the activation energy for bcc and fcc solidification according to classical nucleation theory (CNT) and constructing a boundary line under the condition of ΔG^* (fcc) $= \Delta G^*$ (bcc) [9]. Fair agreement with the experimental results confirms the validity of Eq. (1).

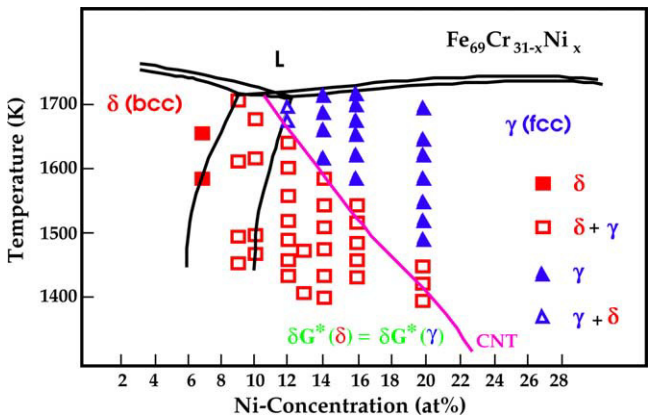

Fig. 2. Phase diagram of the quasi-binary alloy $Fe_{69}Cr_{31-x}Ni_x$ showing the results of undercooling experiments; the symbols classify primary nucleation of the bcc (squares) and the fcc (triangles) phase. The curve denoted by CNT corresponds to the phase-boundary line predicted by classical nucleation theory.

The boundary line strongly depends on the Ni-composition. This clearly demonstrates that undercooling is an important processing parameter enabling the solidification of the metastable bcc-phase owing to its smaller interface energy (σ_{sl}) in the regime of the phase diagram where the fcc-phase is stable. However, because of the small cooling rates in the levitation experiments, the primary metastable bcc-phase cannot be fully conserved during cooling to ambient temperature since it transforms via a solid-state reaction to the stable fcc-phase. The fraction of the metastable phase conserved at room temperature is increased by quenching the solidified sample in a liquid metallic bath [21].

3.3. Primary nucleation of hard magnetic phase in Nd–Fe–B

The intermetallic compound $Nd_2Fe_{14}B_1$ or Φ-phase reveals extraordinary hard magnetic properties, which make Nd–Fe–B alloys suitable for materials as permanent magnets [11,22,23]. Under equilibrium conditions Φ-phase is formed via a peritectic reaction from pro-peritectic γ-Fe. Incomplete peritectic transformation leads to a high amount of soft-magnetic α-Fe that reduces the quality of the magnetic Nd–Fe–B material. Undercooling may open up the possibility of primary solidification of Φ-phase at the expense of γ-Fe. In fact, it is accomplished by melt spinning of thin ribbons [11] and inert gas atomization of very small droplets . Undercooling of bulk melts of Nd–Fe–B alloys suffers from the extreme reactivity of Nd with oxygen forming very stable Nd_2O_3 on the surface of the samples, which act as heterogeneous nucleation sites. Even in electromagnetic levitation experiments, which are conducted under a purified inert gas atmosphere, undercooling of Nd–Fe–B is very limited and solidification starts with the crystallization of the pro-peritectic γ-phase [24]. More recently, a method has been developed to dissolve Nd-oxides by evaporation during electromagnetic levitation of bulk melts [25]. Using this method undercoolings up to 150 K are reached in levitation experiments, sufficiently high to observe three different solidification pathways [12]. The three metastable solidification paths are schematically depicted in Fig. 3 according to experimental investigations on hyperperitectic $Nd_{14}Fe_{79}B_7$ alloy, where the solidification of the intergranular boron-rich and neodymium-rich phases is not included for simplicity. They are only of minor volume fraction. Each solidification mode produces a characteristic microstructure [26]. At small undercooling of $\Delta T = 16$ K (left) dendrites of γ-Fe are primarily formed, followed by peritectic formation of metastable χ-phase $Nd_2Fe_{17}B_x$ ($x \approx 1$) and Φ-phase $Nd_2Fe_{14}B$. $Nd_2Fe_{17}B_x$ decomposes at small undercoolings into γ-Fe and Φ-phase owing to the small cooling rates in levitation experiments (10–100 K/s). At medium undercoolings $\Delta T = 60$ K (middle) the hard-magnetic Φ-phase primarily solidifies from the melt and is the only solidification mode. At large undercoolings $\Delta T = 110$ K (right) metastable χ-phase primarily solidifies and subsequently decomposes into γ-Fe and Φ-phase. To take into account composition besides undercooling, a phase-selection diagram of Nd–Fe–B alloy is constructed, based upon undercooling experiments in electromagnetic levitation [27].

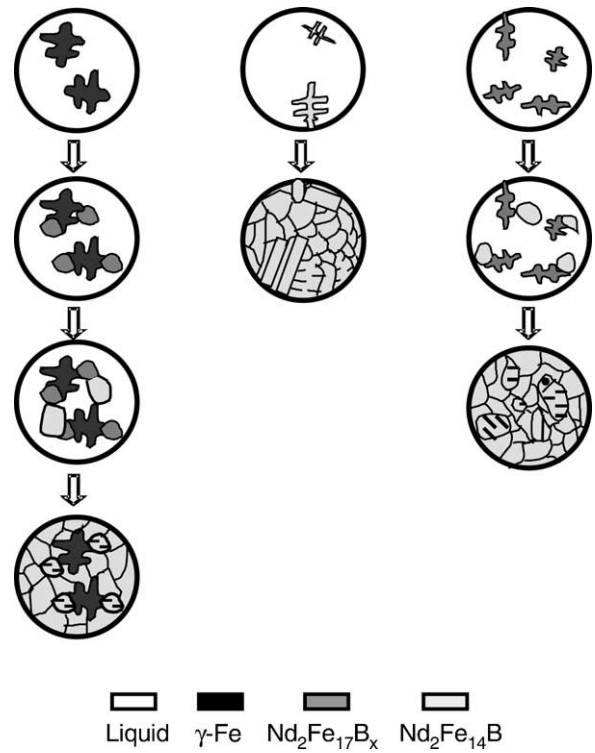

Fig. 3. Schematic summary of three different solidification pathways of hyperperitectic $Nd_{14}Fe_{79}B_7$ alloy observed at three different undercoolings. Small undercooling: $\Delta T = 16$ K (left): primary γ-Fe formation, subsequent peritectic formation of $Nd_2Fe_{17}B_x$ (χ-phase) and $Nd_2Fe_{14}B$ (Φ-phase); medium undercooling: $\Delta T = 60$ K (middle): primary $Nd_2Fe_{14}B$ formation; large undercooling: $\Delta T = 110$ K (right): primary $Nd_2Fe_{17}B_x$ formation, decomposition of $Nd_2Fe_{17}B_x$ into γ-Fe and (Φ-phase).

3.4. Direct observation of phase-selection using synchrotron radiation

The conclusions of the aforementioned undercooling experiments are based upon the analysis of the thermodynamics. They do not directly evidence the solidification pathways during rapid solidification of the undercooled melt. After cooling to ambient temperature primarily formed metastable phase has been transformed either by remelting during recalescence or by solid-state transformation. X-ray diffraction on the as solidified samples at room temperature is therefore not helpful to detect the metastable solid product. The "lifetime" τ of the metastable phase is too short to record a diffraction pattern by conventional diffraction and to identify the crystallographic phase unambiguously. The situation is changed if high-intensity synchrotron radiation is considered.

The electromagnetic levitation technique to undercool bulk samples was combined with energy dispersive X-ray diffraction (EDXD) to record in situ complete diffraction patterns during crystallization of stable and metastable phases in undercooled metallic melts. The EDXD experiments were performed at the white beam Laue beamline ID 09 at the European Synchrotron Radiation Facility (ESRF) in Grenoble, France. Unfocused radiation from a wiggler with an energy range from 7 to 148 keV and an energy dispersive germanium detector at a constant scattering angle of 5° was used. The experimental set up is described in detail elsewhere [13]. The minimum integration time to observe a spectrum of reasonable quality was 0.5 s, which is the limiting factor for observing time-resolved phase-selection in undercooled samples.

While the "lifetime" τ of metastable bcc structures in Fe–Ni and Fe–Ni–Cr alloys is within milliseconds [8,9] considerably longer times were reported for some Ni–V alloys ranging from several milliseconds for Ni-rich alloys and some seconds for V-rich alloys [10]. In the composition range between 42 and 63 at.% V, Ni–V shows a phase competition between three different crystallographic phases, bcc, fcc, and the intermetallic σ'-phase. The σ'-phase shows a polytetrahedral short range order in its unit cell similar to that of quasicrystal forming alloys. Since phases of tetrahedral short range order should possess a small interfacial energy, ΔG^* is expected to be small [17]. In case of σ'-phase in Ni–V an interfacial energy has been calculated with $\alpha = 0.36$ [28]. The crystallisation of the σ'-phase, therefore, should be favored compared with the phases of cubic structure. Surprisingly, this is in contrast to the experiments.

As an example, Fig. 4 shows a temperature–time profile (cf. inset) and four diffraction spectra recorded during undercooling and solidification of a levitated $Ni_{41}V_{59}$ sample. In the temperature–time profile a double recalescence is observed. The first recalescence peak sets in at an undercooling of $\Delta T = 50$ K ($T_N = 1493$ K) and T rises up to $T = 1514$ K being below the liquidus temperature of the stable σ'-phase, $T_l^{\sigma} = 1543$ K. After 57 s, the remaining undercooled liquid crystallizes in a subsequent step and T rises up to 1490 K. This temperature is below the liquidus temperatures of both the metastable bcc-phase, $T_l^{bcc} = 1533$ K, and the stable σ'-phase, T_l^{σ}. Hence, the determination of the post-recalescence temperatures does not allow for the discrimination of the phases solidified during the two different recalescence events.

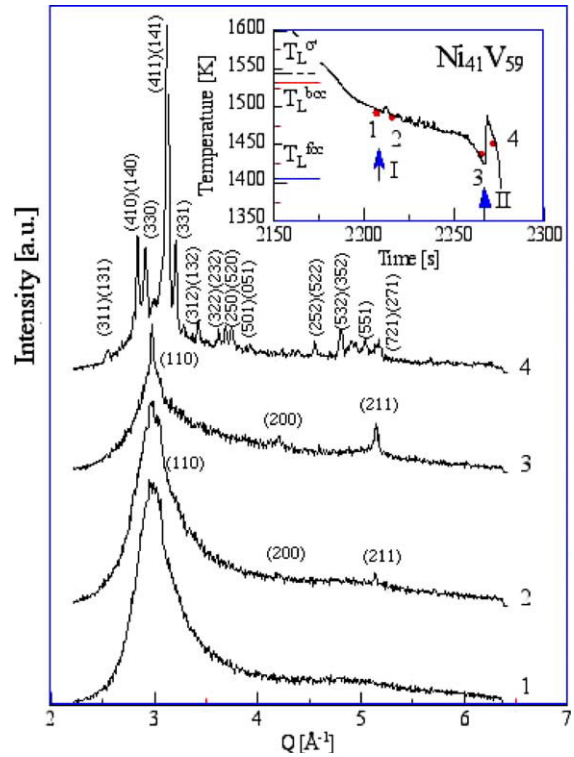

Fig. 4. Energy-dispersive X-ray diffraction spectra recorded during undercooling and solidification of a levitated $Ni_{41}V_{59}$ at a constant diffraction angle of $2\theta = 5°$. A part of the undercooled melt (spectrum 1) primarily crystallizes into the metastable bcc-phase (spectra 2 and 3) followed by the crystallization of the intermetallic σ'-phase (spectrum 4). The bcc-phase disappears by transforming during the second solidification period. The inset marks the temperature and time during cooling at which the various spectra are recorded. The arrows denoted by I and II mark the two recalescence events of primary bcc and secondary σ' crystallization [14].

The crystalline structures of the solidifying phases were directly identified by the diffraction pattern immediately recorded after the two recalescence events as shown in Fig. 4. The first spectrum has been taken on the undercooled liquid at $T = 1495$ K, the second one directly after the first recalescence event and the third one 54 s after that event. Spectrum 2 clearly reveals that after the first recalescence event (denoted by I in Fig. 4) a part of the undercooled melt crystallized into metastable bcc-phase which grows continuously upon further cooling as shown by spectrum 3. In both spectra the Bragg peaks are identified as the (1 1 0), (2 0 0), and (2 1 1) reflections of the metastable bcc-phase. In a second reaction the remaining liquid crystallizes into the stable σ'-phase as indicated by spectrum 4 of Fig. 4. The Bragg-peaks of the bcc-phase disappear during the second solidification process,

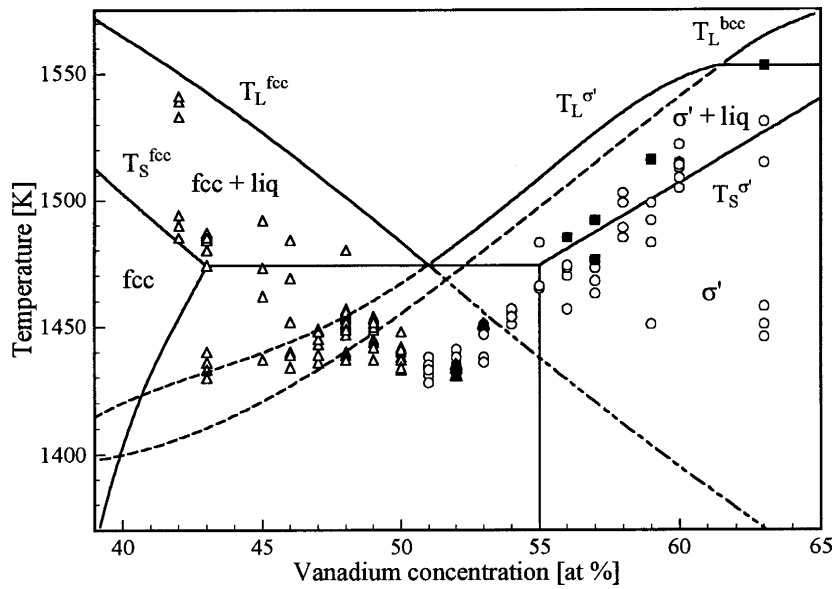

Fig. 5. Phase-selection diagram of Ni–V. The different symbols mark the primarily solidified phases as a function of composition and nucleation temperature, open triangles: fcc-phase, closed triangles: fcc-phase + σ′-phase, open circles: bcc-phase and closed squares: σ′-phase. The solid lines give the equilibrium liquidus and solidus temperatures while dash, dotted, and dash-dotted lines represent the metastable extensions of liquidus temperatures of bcc, σ′ and fcc-phase, respectively.

indicating that the metastable bcc-phase transforms during the temperature rise in the second recalescence event.

The analysis of equivalent X-ray diffraction experiments performed on samples of different composition and solidified at various levels of undercooling leads to the development of a phase-selection diagram for Ni–V shown in Fig. 5 [14]. The equilibrium liquidus (T_l) and solidus temperatures (T_s) of fcc, bcc, and σ'-phases are shown as solid lines and the extensions of T_l for all phases into the metastable region of the undercooled melt are represented by dashed (bcc), dotted (fcc) and dash-dotted (σ') lines. The symbols denote the primarily solidified phase as a function of composition and nucleation temperature: fcc-phase (open triangles), bcc-phase (open circles), fcc-phase + σ'-phase (closed triangles), and σ'-phase (closed squares).

The comparison of the experimental results with the calculated phase diagram shows that the formation of the metastable bcc-phase is observed in the concentration range between 51 and 60 at.% V at nucleation temperatures below the metastable extension of bcc liquidus, i.e. in a temperature and concentration regime in which a driving force for the solidification of bcc-phase does exist: $\Delta G = G(\text{bcc}) - G(\text{l}) < 0$ (G(bcc) and G(l): Gibbs free energy of bcc-solid and liquid). This is a necessary but not a sufficient condition for the formation of metastable bcc-phase. Usually, ΔG is largest for the stable phase in comparison with any metastable solid [29]. Crystallization of metastable solids can be understood, if nucleation processes are taken into account. The σ'-phase is a Frank–Kasper phase with a complex polytetrahedral structure [30]. The NaCl type structure of the VO shows more similarity to the cubic structure of bcc and fcc-phase than to the polytetrahedral structure of the σ'-phase. Consequently, $f(\theta)$ for heterogeneous nucleation on Ni and V-oxides is expected to be larger for the σ'-phase than for the cubic bcc-phase, hence $\Delta G^*(\text{bcc}) < \Delta G^*(\sigma')$, in qualitative agreement with the experimental findings. These experiments show the importance of metal-oxides on the solidification behavior. In the case of Ni–V vanadium-oxides are formed, which are stable at elevated temperatures above the liquidus temperatures of the metallic Ni–V samples. They show a cubic crystalline structure being more similar to the fcc or bcc-phases than to σ'-phase. Therefore, the catalytic potency for heterogeneous nucleation as represented by the factor $f(\theta)$ due to the vanadium oxides is more pronounced for the nucleation of the cubic phases than of the intermetallic σ'-phase. The metal oxides of Ni and V therefore stimulate the heterogeneous crystal nucleation of fcc and bcc-phase in preference of the σ'-phase.

4. Summary and conclusions

It has been shown that the electromagnetic levitation technique is a very efficient tool to undercool bulk melts far below their melting temperatures. A freely suspended drop gives direct access to observe the primary formation of metastable phases during rapid crystallization of deeply undercooled melts by using proper diagnostic means. High-speed temperature recording of the rapid temperature rise during recalescence allows for the discrimination of the sequence of the formation of different crystallographic phases. It is demonstrated that controlled seeding of a

metastable phase in levitation-processed samples can be achieved by using an external nucleation trigger of structure similar to the metastable phase to be nucleated. A complete phase-selection diagram of isopleth $Fe_{69}Ni_{31-x}Cr_x$ steel alloy is presented showing both composition and undercooling range where metastable ferrite is primarily solidified. Even of more practical importance it could be shown that primary crystallization of hard magnetic Φ-phase ($Nd_2Fe_{14}B$) intermetallic phase is possible in hyperperitectic $Nd_{14}Fe_{79}B_7$ alloy through undercooling the liquid prior to solidification by moderate amounts. In such a way, the crystallization of soft-magnetic γ-Fe phase is almost completely suppressed. The results on the primary formation of metastable solids are based upon a thermal analysis of the measured recalescence profiles. But the validity of this analysis is evidenced by X-ray diffraction experiments using high-intensity synchrotron radiation at the ESRF Grenoble. Such, complete diffraction spectra were recorded in a time interval less than 1 s leading to a data base for constructing experimentally determined phase-selection diagrams as proven by a set of measurements on binary Ni–V alloy.

Synchrotron radiation is a very attractive method to observe phase-selection processes during rapid solidification of undercooled melts. Diffraction experiments are in progress to unambiguously determine the sequence of phase formation in undercooled melts of Nd–Fe and Nd–Fe–B alloys [31]. The formation of crystallographic phases in undercooled melts requires atomic diffusion, which in turn is strongly influenced by fluid flow in the liquid driven by natural convection and electromagnetic stirring of levitated drops. A change of fluid flow therefore may be used to control phase-selection during solidification. First experiments in reduced gravity during parabolic flight campaigns using a facility for containerless processing in space (MSL–EML)[1] show that under such conditions the critical undercoolings for primary solidification of metastable phases is essentially changed [32].

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[1] MSL–EML is a multiuser facility which is presently developed by the German aerospace Center DLR and the European Space Agency ESA for the accommodation on board of the International Space Station.

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